Experiment 4: The Sol-Gel Technique – Lab Report

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1) Aim

To synthesize SiO_2 nanopowder by the sol-gel method, coat it by dip and spin coating, and examine the effect of SiO_2 content on the contact angle and viscosity of PVA solutions.

2) Materials and Devices

TEOS, ethanol, distilled water, NH_4F, NH_4OH, PVA, Triton X-100; magnetic stirrer, balance, drying oven, furnace, laser, spin coater, FTO/ITO glass, Ostwald viscometer.

3) Experimental Procedure

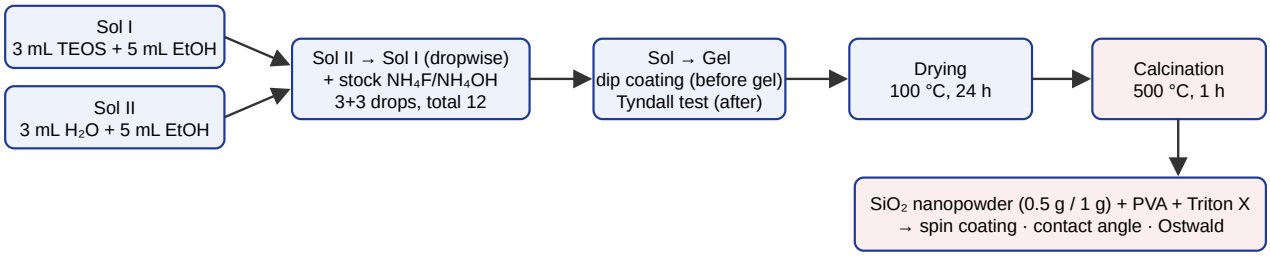


Figure 1. Flow chart of the experiment.

- Solution II (3 mL water + 5 mL ethanol) was added dropwise to Solution I (3 mL TEOS + 5 mL ethanol) under stirring; hydrolysis started (≡Si–OR + H_2O → ≡Si–OH + ROH).
- Stock catalyst (1.9 g NH_4F + 23 mL NH_4OH in 100 mL water [1]) was added 3 drops at a time until a coating formed on a dipped glass slide (12 drops in total).
- Before gelation a glass slide was dip-coated. After gelation a laser was passed through water and the gel.
- The gel was dried at 100 °C for 24 h and calcined at 500 °C for 1 h, and weighed after each step.
- PVA suspensions with 0.5 g and 1 g SiO_2 were spin-coated (3 drops, 500/800 rpm, 5/10 s). One drop of water was placed on each coating, and the viscosities were measured with the Ostwald viscometer (5 mL, water as reference).

4) Results and Discussion

4.1) Percent weight loss

Step	Mass (g)	Loss (%)
Wet gel	10.900	–
Dried (100 °C, 24 h)	0.576	94.72
Calcined (500 °C, 1 h)	0.394	31.60
Total		**96.39**

$$Loss\ (\%) = (m_{before} - m_{after}) / m_{before} \times 100$$

Drying: (10.9 − 0.576)/10.9 × 100 = 94.72 %
Calcination: (0.576 − 0.394)/0.576 × 100 = 31.60 %
Total: (10.9 − 0.394)/10.9 × 100 = 96.39 %

Drying removed the water and ethanol in the gel; calcination removed residual organics and OH groups.

4.2) Tyndall effect

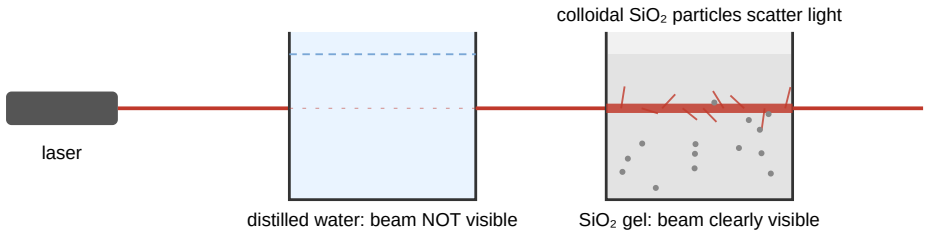
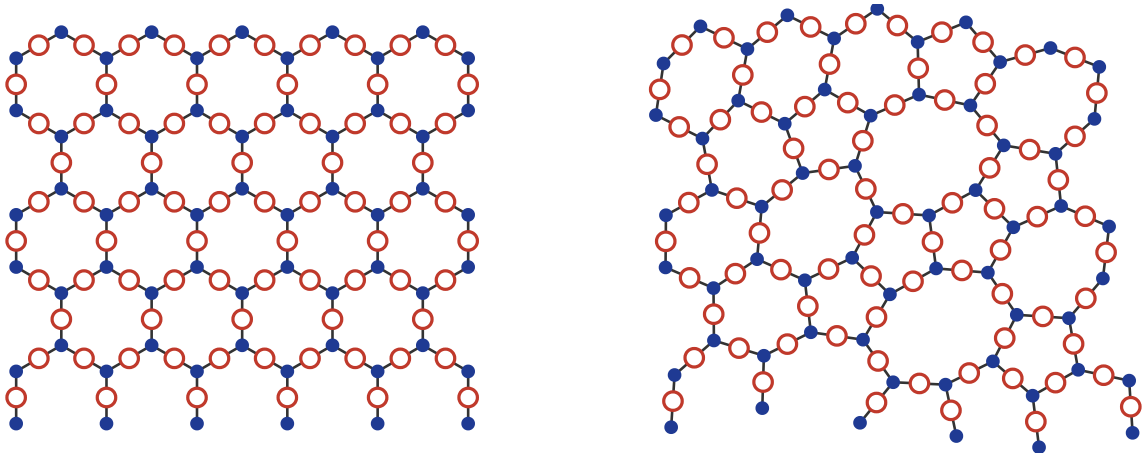


Figure 2. Observation of the Tyndall effect.

The beam was invisible in water but visible in the gel, because the colloidal SiO_2 particles scatter light. This shows that the gel is a colloid.

4.3) Crystalline and amorphous SiO_2



(a) Crystalline SiO_2 *(b) Amorphous SiO_2*

● Si ○ O *Figure 3. 2-D schematic of the SiO_4 network.*

Property	Crystalline (quartz)	Amorphous (vitreous)
Order	Long-range, periodic	Only short-range, random rings
Density	2.65 g/cm³	~2.2 g/cm³
Melting	Sharp melting point	Softens gradually (T_g)
XRD / optics	Sharp peaks, anisotropic	Broad hump, isotropic
Chemical	Few Si–OH, less reactive	Many Si–OH, high surface area, more reactive

Our powder is amorphous, because it formed by random condensation and was heated only to 500 °C. **Applications:** fillers for polymers and rubber, catalyst supports, adsorbents, drug delivery, coatings, polishing slurries, food and cosmetic additives.

4.4) Contact angle and superhydrophobic surface

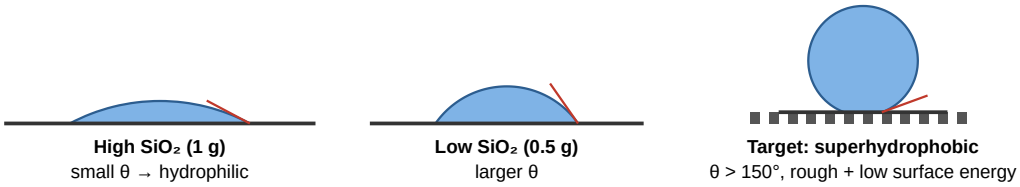


Figure 4. Observed drop shapes (schematic) and the target surface.

The contact angle decreased as SiO_2 increased, so the surface became more hydrophilic, because silica carries polar Si–OH groups. To make it superhydrophobic, the Si–OH groups are replaced with a fluoroalkylsilane (FAS) [2]:

$$\equiv Si–OH + (C_2H_5O)_3Si–CH_2CH_2(CF_2)_nCF_3 \rightarrow \equiv Si–O–Si–CH_2CH_2(CF_2)_nCF_3 + C_2H_5OH$$

The fluorinated groups lower the surface energy, and the SiO_2 nanoparticles give roughness that traps air under the drop. Brassard et al. spin-coated FAS-treated silica nanoparticles in this way and obtained a contact angle of 151°, so rain drops roll off [2].

4.5) Spin coating: effect of rpm and time

Higher rpm → thinner film: the final thickness follows $h \propto \omega^{-1/2}$ [3]; going from 500 to 800 rpm gives a ~20 % thinner film. **Longer time → thinner only in the first seconds:** after that, solvent evaporation stops the flow and the thickness stays constant [3]. This is why 5 s and 10 s gave the same coating.

4.6) Viscosity

Liquid	ρ (g/cm³)	t (s)	η (mPa·s)
Water (ref.)	1.000	1.45	0.89
Low SiO_2	1.02586	8.06	**5.08**
High SiO_2	1.02664	8.34	**5.26**

$$\eta = \eta_0 \cdot (\rho \cdot t) / (\rho_0 \cdot t_0)$$

ρ = m/V: 5.1293/5 = 1.02586 ; 5.1332/5 = 1.02664 g/cm³
Low: 0.89 × (1.02586 × 8.06)/(1 × 1.45) = 5.08 mPa·s
High: 0.89 × (1.02664 × 8.34)/(1 × 1.45) = 5.26 mPa·s

The viscosity increased with SiO_2 content, because more particles increase internal friction and bond to PVA chains through hydrogen bonds.

5) References

1. Experiment 4: The Sol-Gel Technique, MSE4121 Material Production Laboratory Experimental Booklet, Yıldız Technical University, 2026.
2. Brassard, J.-D., Sarkar, D. K., Perron, J., "Synthesis of monodisperse fluorinated silica nanoparticles and their superhydrophobic thin films", *ACS Appl. Mater. Interfaces* 3 (2011) 3583–3588. doi:10.1021/am2007917
3. Meyerhofer, D., "Characteristics of resist films produced by spinning", *J. Appl. Phys.* 49 (1978) 3993–3997. doi:10.1063/1.325357